MIL-I-60529(MU)
28 April 1967
SUPERSEDING
FA-PD-MI-2587, Rev. 0
22 March 1963

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, DELAY, M89, ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, M89, Assembly, loaded with Cartridge, Initiator, Delay, M175, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MS-28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

DRAWINGS

Munitions Command

F10522698	- Initiator, Cartridge Actuated, Delay, M89, Assembly
C8846785	- Container, Ammunition, Fiber, M459 for Initiator, Delay, Cartridge Actuated

FSC 1377

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DRAWINGS (Cont'd)

C8846786 - Box, Packing, Ammunition, for Initiator, Delay, Cartridge Actuated in Fiber Container, M459

IEL-10522698 - Inspection Equipment List for Initiator, Cartridge Actuated, Delay, M89, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Firing pin release.- The firing pin of the assembled initiator, less safety pin and cartridge, shall release from the cocked position when the application of an axial load of 15 to 30 pounds has withdrawn the initiator pin .370 inches, minimum, and .425 inches, maximum.

3.3 Initiator pin security.- Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permenant deformation of the safety pin.

3.4 Breakaway torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.5 Residual magnetism.- The initiator shall not defect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.6 Functional.

3.6.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be produced at the end of 15 feet of aircraft hose, size No. 4, as specified in MS-28741.

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Table I

Temperature	Pressure (PSI) Min.	Max.	Ignition Delay (See 6.2)
-65° to -70°F	500	1500	6.0 seconds, Max.
65° to 75°F	575	1500	5.0 ± 0.30 seconds
160° to 165°F	625	1500	4.0 seconds, Min.

3.6.2 Carbon monoxide gas leakage.- Initiators loaded with accepted cartridges and temperature conditioned at minus 65-5 degrees, 70+5 degrees and 165+5 degrees F., when fired into a sealed total volume of .70 cubic inches shall not produce carbon monoxide gas leakage greater than 20.0 cubic centimeters total. Volume of gas leakage shall be recorded at 10 minutes and 60 minutes after firing.

3.6.3 The initiator shall not fail to fire.

3.6.4 The initiator shall function without mechanical failure. (See 6.1)

3.7 Workmanship.- Requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Initiator, Delay, M175, Dwg. D10522660, covering a detail of Dwg. F10522698.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual

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4.2.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Seal, Initiator, Dwg. C10522692, covering a detail of Dwg. F10522697.

CRITICAL	
None defined	
MAJOR	
101. Diameter, outside, min.	Gage
102. Width	Gage
103. Thickness	Gage
104. Material defective	Visual
105. Flash	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter, outside, max.	Gage
202. Width of taper	Gage
203. Width to end of taper	Gage
204. Radius or angle missing	Visual

4.2.2.3 Assembly, Dwg. F10522698, less all drawings except C10520410, C10520411 and MS 19060-3.

CRITICAL	
1. Firing pin protrusion	Gage
2. Steel ball missing	Gage
3. Striker hole in firing pin housing obstructed	Visual

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4.2.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Assembly, Dwg F10522698, less Dwgs D10520335, D8597928, C10520414, and B8359536.

CRITICAL	
1. Firing pin not cocked	Manual & Visual
2. Any part missing	Visual
3. Improper assembly	Visual
4. Foreign matter on seal	Visual
MAJOR	
101. Safety pin loose or deformed	Visual
102. Burr	Visual
103. Foreign matter unless otherwise classified	Visual
104. Seal not lubricated	Visual

4.2.2.5 Assembly, Dwg F10522698, less all drawings except D8597928, C10520414 and D10520335.

CRITICAL	
1. Cartridge not properly seated	Visual
2. Filter missing or insecure	Visual
3. Nylon patch missing or improperly applied	Visual
4. Burrs or nicks on edge of chamber mouth	Visual
5. Toolmarks, scratches or nicks on cartridge seat of chamber	Visual
MAJOR	
101. Burr, except as otherwise classified	Visual
102. Foreign matter	Visual

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4.2.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
None defined	
4.2.2.6 Assembly, Dwg F10522698.	
CRITICAL	
1. Cartridge face not seated against base of firing pin housing	X-ray
2. Any part missing	X-ray
3. Improper assembly	X-ray
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Set screw missing or insecure	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
201. Thread sealing compound missing from set screws	Visual
202. Shipping cap missing	Visual
203. Protective finish inadequate or defective	Visual
4.2.2.7 Container, Ammunition, Fiber, M459, Dwg C8846785.	
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual

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4.2.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab,min.	Visual
208. Color of tape incorrect	Visual

4.2.2.8 Box, Packing, Ammunition Dwg C8846786.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual & Visual
203. Strapping missing, loose, broken or improperly assembled	Manual & Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3:

4.2.3.1 Firing pin release.- The initiator pin of each subassembly, consisting of all parts except the safety pin, cartridge and chamber, shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Initiator pin security.- The initiator pin of each subassembly, consisting of all parts except the chamber and cartridge, shall be subjected to the initiator pin security tests. Failure of any subassembly to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

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4.2.3.3 Breakaway torque.- Each initiator shall be subjected to the breakaway torque application. Failure of any initiator assembly to comply with the requirements of 3.4 shall be cause for rejection of the assembly.

4.2.3.4 Residual magnetism.- Each initiator shall be subjected to the test for excessive residual magnetism. Failure of any initiator to comply with the requirements of 3.5 shall be cause for rejection of the initiator.

4.2.3.5 Functional.- The following tests shall be performed by the Government:

4.2.3.5.1 Ballistics.- Thirty units shall be tested to determine compliance with the requirements of 3.6.1, 3.6.3 and 3.6.4. The units shall be temperature conditioned as provided in Table III and 4.3.6.1.1. Failure of any unit to comply with the requirements of 3.6.1, 3.6.3 and 3.6.4 shall be cause for rejection of the lot.

Table III

Temperature	Sample Size
-65° to -70°F	10
65° to 75°F	10
160° to 165°F	10

4.2.3.5.2 Carbon monoxide gas leakage.- Fifteen units shall be tested to determine compliance with 3.6.2. The units shall be temperature conditioned as provided in Table IV and 4.3.5.1.1. Failure of any unit to comply with the requirements of 3.6.2 shall be cause for rejection of the lot.

Table IV

Temperature	Sample Size
-65° to -70°F	5
65° to 75°F	5
160° to 165°F	5

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10522698.

4.3 Test methods and procedures.

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4.3.1 Firing pin release.- The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the fixture which registers the amount of pull.

4.3.2 Initiator pin security.- The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.3.3 Breakaway torque.- A torque wrench set at the torque requirement specified in 3.4 shall be used in conjunction with the fixture to determine the adequacy of the joint. The torque shall be applied to the cap.

4.3.4 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects, by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.5 Functional.

4.3.5.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample unit shall be re-conditioned in accordance with 4.3.5.1.1.

4.3.5.1.1 Temperature conditioning.- A standard M89 initiator, loaded with a dummy M175 cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.3.5.2 Carbon monoxide gas leakage.- The test sample shall be rigidly supported on the upright member of the fixture. The sealed actuating rod shall be attached to the initiator pull pin. A length of flared stainless steel tubing shall be attached to the initiator outlet. A length of MS28741-4 flexible hose, closed at one end with a locked shut cap, shall be connected to the stainless steel tubing. The combined volumes of the tubing

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4.3.5.2 (Cont'd)

and flexible hose shall simulate the functional volume specified in 3.6.2. The stainless steel tubing length shall be such that the connection between the tubing and the flexible hose will be isolated from the bell jar interior. One connection only shall be permitted in the interior of the bell jar. The safety pin shall be removed from the test sample and the bell jar positioned and sealed to the base plate. A protective wire mesh cover shall be placed over the bell jar. The total set-up time shall not exceed 10 minutes, maximum, before firing. When this time limit is exceeded, the sample unit shall be re-conditioned at the original temperature. Immediately prior to firing, the 3-way petcock shall be aligned to assure a direct flow into the rubber bulb. After firing, the bulb shall be compressed and released several times to assure positive circulation of all gases within the bell jar. At 10 minutes after each firing, the petcock shall be re-oriented to provide alignment with the MSA Dosimeter for carbon monoxide measurement. This procedure shall be repeated at 60 minutes after each firing. A gas sample shall be withdrawn for chromatograph analysis from the 1st, 3rd, and 5th units at each temperature at both 10 and 60 minutes after firing. In the event that the carbon monoxide leakage of the 2nd or 4th unit exceeds the maximum scale of the MSA Dosimeter, a gas sample shall be withdrawn at 60 minutes after firing for chromatograph analysis.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Dwgs C8846785 and C8846786.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 "Ignition delay" is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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